

# Title of the Thesis

## Optional Subtitle

by

**Forename Surname**

February, 2024

*A thesis submitted to the  
Graduate School  
of the  
Institute of Science and Technology Austria  
in partial fulfillment of the requirements  
for the degree of  
Master of Science*

Committee in charge:

Name Surname, Supervisor

Name Surname

Name Surname





The thesis of Forename Surname, titled *Title of the Thesis*, is approved by:

**Supervisor:** Name Surname, ISTA, Klosterneuburg, Austria

Signature: \_\_\_\_\_

**Co-supervisor:** Name Surname, Institution, Place, Country

Signature: \_\_\_\_\_

**Committee Member:** Name Surname, ISTA, Klosterneuburg, Austria

Signature: \_\_\_\_\_

Signed page is on file



© by Forename Surname, February, 2024

CC BY-NC-SA 4.0 The copyright of this thesis rests with the author. Unless otherwise indicated, its contents are licensed under a Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International License. Under this license, you may copy and redistribute the material in any medium or format. You may also create and distribute modified versions of the work. This is on the condition that: you credit the author, do not use it for commercial purposes and share any derivative works under the same license.

ISTA Master's Thesis, ISSN: 2791-4585

ISBN: 978-0-00000-000-0

I hereby declare that this thesis is my own work and that it does not contain other people's work without this being so stated; this thesis does not contain my previous work without this being stated, and the bibliography contains all the literature that I used in writing the dissertation.

I accept full responsibility for the content and factual accuracy of this work, including the data and their analysis and presentation, and the text and citation of other work.

I declare that this is a true copy of my thesis, including any final revisions, as approved by my thesis committee, and that this thesis has not been submitted for a higher degree to any other university or institution.

I certify that any republication of materials presented in this thesis has been approved by the relevant publishers and co-authors.

Signature: \_\_\_\_\_

Forename Surname  
February, 2024

Signed page is on file



# Abstract

Summarize the gist of your thesis in one page.

# Acknowledgements

Mention grants if you were funded. Consider Scientific Service Units (SSUs) at ISTA and other staff if you got their support.



# About the Author

Include paragraph of text summarizing your educational background, your academic profile (e.g. relevant research projects or work experience, list of publications), to give readers a snapshot idea of you as a person/academic. It is not meant to be a CV in the sense that it is not intended for job applications. This means things like your email address, your nationality, your current address, etc., should be left out.

The following is an entirely fictional blurb to give you an idea:  
Jane Doe completed a BSc in Natural Sciences at the University of Cambridge and an MSc in Cell Biology at the University of Edinburgh before joining ISTA in September 2016. Her main research interests include evolution of genetic traits and environmental pressures on evolution. She worked on the research project “Mathematical Modeling of Evolutionary Constraints” with the Abel group at the University of Vienna in the summer of 2014, and published these results in the high-impact journal *Evolutionary Biology*. During her PhD studies, Jane also presented her research results at the EvoBIO conference in Parma in 2018 and helped develop the tool EVOBOT for modeling evolutionary constraints, currently hosted by ISTA.

# List of Collaborators and Publications

Specify all co-authors and collaborator contributions to this thesis. See the Guidelines for Thesis Submission for details.

List all publications that appear in this thesis.

Use the `\bibentry` command to produce a full citation:

Alan Mathison Turing. On computable numbers, with an application to the Entscheidungsproblem. *J. of Math*, 58:345–363, 1936

# Table of Contents

|                                                      |             |
|------------------------------------------------------|-------------|
| <b>Abstract</b>                                      | <b>vii</b>  |
| <b>Acknowledgements</b>                              | <b>viii</b> |
| <b>About the Author</b>                              | <b>ix</b>   |
| <b>List of Collaborators and Publications</b>        | <b>x</b>    |
| <b>Table of Contents</b>                             | <b>xi</b>   |
| <b>List of Figures</b>                               | <b>xii</b>  |
| <b>List of Tables</b>                                | <b>xii</b>  |
| <b>List of Algorithms</b>                            | <b>xii</b>  |
| <b>List of Terms</b>                                 | <b>xii</b>  |
| <b>List of Abbreviations</b>                         | <b>xiii</b> |
| <b>1 Introduction</b>                                | <b>1</b>    |
| <b>2 Additional Chapter</b>                          | <b>3</b>    |
| <b>3 Introduction to L<sup>A</sup>T<sub>E</sub>X</b> | <b>5</b>    |
| 3.1 Installation . . . . .                           | 5           |
| 3.2 Editors . . . . .                                | 5           |
| 3.3 Compilation . . . . .                            | 6           |
| 3.4 Basic Functionality . . . . .                    | 6           |
| 3.5 Bibliography . . . . .                           | 8           |
| 3.6 Table of Contents . . . . .                      | 9           |
| 3.7 Acronyms / Glossary / Index . . . . .            | 9           |
| 3.8 Tips . . . . .                                   | 9           |
| 3.9 Resources . . . . .                              | 10          |
| <b>Bibliography</b>                                  | <b>13</b>   |
| <b>A Example Appendix Chapter</b>                    | <b>15</b>   |

## List of Figures

|     |                                                                                   |   |
|-----|-----------------------------------------------------------------------------------|---|
| 3.1 | Optional caption for the figure list (often used to abbreviate long captions) . . | 8 |
|-----|-----------------------------------------------------------------------------------|---|

## List of Tables

|     |                                                                                                                                                                                     |    |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 3.1 | T <sub>E</sub> X/L <sup>A</sup> T <sub>E</sub> X distributions for different operating systems. Recommended choice is in <b>bold</b> . . . . .                                      | 5  |
| 3.2 | Compilation steps for this document. The following abbreviations were used: table of contents (toc), list of figures (lof), list of tables (lot), list of algorithms (loa). . . . . | 6  |
| 3.3 | L <sup>A</sup> T <sub>E</sub> X table example with shortened caption for the list of tables . . . . .                                                                               | 7  |
| 3.4 | Anonymization of PDF documents. . . . .                                                                                                                                             | 10 |

## List of Algorithms

|     |                        |   |
|-----|------------------------|---|
| 3.1 | Gauss-Seidel . . . . . | 8 |
|-----|------------------------|---|

## List of Terms

**editor** A text editor is a type of program used for editing plain text files.. 5

# List of Abbreviations

**CTAN** Comprehensive TeX Archive Network. 10

**FAQ** Frequently Asked Questions. 10

**PDF** Portable Document Format. xii, 6, 9, 10

**SVN** Subversion. 9

**WYSIWYG** What You See Is What You Get. 9



# CHAPTER 1

## Introduction

Enter  
your text  
here.





# CHAPTER 2

## Additional Chapter

Enter  
your text  
here.



## Introduction to $\LaTeX$

Since  $\LaTeX$  is widely used in academia and industry, there exists a plethora of freely accessible introductions to the language. Reading through the guide at <https://en.wikibooks.org/wiki/LaTeX> serves as a comprehensive overview of most of the functionality and is highly recommended before starting with a thesis in  $\LaTeX$ .

### 3.1 Installation

A full  $\LaTeX$  distribution consists not only of the binaries that convert the source files to the typeset documents but also of a wide range of packages and their documentation. Depending on the operating system, different implementations are available as shown in Table 3.1. **Due to the large number of packages that are in everyday use and due to their high interdependence, it is paramount to keep the installed distribution up to date.** Otherwise, obscure errors and tedious debugging ensue.

### 3.2 Editors

A multitude of  $\TeX$  editors are available differing in their editing models, their supported operating systems, and their feature sets. A comprehensive overview of editors can be found on the Wikipedia page [https://en.wikipedia.org/wiki/Comparison\\_of\\_TeX\\_editors](https://en.wikipedia.org/wiki/Comparison_of_TeX_editors).  $\TeX$ studio (<http://texstudio.sourceforge.net/>) is recommended. Most editors support synchronization of the generated document and the  $\LaTeX$  source by `Ctrl`-clicking either on the source document or the generated document.

| Distribution | Unix       | Windows    | MacOS      |
|--------------|------------|------------|------------|
| TeX Live     | <b>yes</b> | yes        | (yes)      |
| MacTeX       | no         | no         | <b>yes</b> |
| MikTeX       | (yes)      | <b>yes</b> | yes        |

Table 3.1:  $\TeX$ / $\LaTeX$  distributions for different operating systems. Recommended choice is in **bold**.

| Description |                                                                                                     |
|-------------|-----------------------------------------------------------------------------------------------------|
| 1           | Scan for refs, toc/lof/lot/loa items and cites                                                      |
| 2           | Build the bibliography                                                                              |
| 3           | Link refs and build the toc/lof/lot/loa                                                             |
| 4           | Link the bibliography                                                                               |
| 5           | Build the glossary                                                                                  |
| 6           | Build the acronyms                                                                                  |
| 7           | Build the index                                                                                     |
| 8           | Link the glossary, acronyms, and the index                                                          |
| 9           | Link the bookmarks                                                                                  |
| Command     |                                                                                                     |
| 1           | <code>pdflatex.exe example</code>                                                                   |
| 2           | <code>bibtex.exe example</code>                                                                     |
| 3           | <code>pdflatex.exe example</code>                                                                   |
| 4           | <code>pdflatex.exe example</code>                                                                   |
| 5           | <code>makeindex.exe -t example.glg -s example.ist</code><br><code>-o example.gls example.glo</code> |
| 6           | <code>makeindex.exe -t example.alg -s example.ist</code><br><code>-o example.acr example.acn</code> |
| 7           | <code>makeindex.exe -t example.ilg -o example.ind example.idx</code>                                |
| 8           | <code>pdflatex.exe example</code>                                                                   |
| 9           | <code>pdflatex.exe example</code>                                                                   |

Table 3.2: Compilation steps for this document. The following abbreviations were used: table of contents (toc), list of figures (lof), list of tables (lot), list of algorithms (loa).

### 3.3 Compilation

Modern editors usually provide the compilation programs to generate Portable Document Format (PDF) documents and for most L<sup>A</sup>T<sub>E</sub>X source files, this is sufficient. More advanced L<sup>A</sup>T<sub>E</sub>X functionality, such as glossaries and bibliographies, needs additional compilation steps, however. It is also possible that errors in the compilation process invalidate intermediate files and force subsequent compilation runs to fail. It is advisable to delete intermediate files (`.aux`, `.bbl`, etc.), if errors occur and persist. All files that are not generated by the user are automatically regenerated. To compile the current document, the steps as shown in Table 3.2 have to be taken.

### 3.4 Basic Functionality

In this section, various examples are given of the fundamental building blocks used in a thesis. Many L<sup>A</sup>T<sub>E</sub>X commands have a rich set of options that can be supplied as optional arguments. The documentation of each command should be consulted to get an impression of the full spectrum of its functionality.

### 3.4.1 Floats

Two main categories of page elements can be differentiated in the usual  $\text{\LaTeX}$  workflow: (i) the main stream of text and (ii) floating containers that are positioned at convenient positions throughout the document. In most cases, tables, plots, and images are put into such containers since they are usually positioned at the top or bottom of pages. These are realized by the two environments `figure` and `table`, which also provide functionality for cross-referencing (see Table 3.3 and Figure 3.1) and the generation of corresponding entries in the list of figures and the list of tables. Note that these environments solely act as containers and can be assigned arbitrary content.

### 3.4.2 Tables

A table in  $\text{\LaTeX}$  is created by using a `tabular` environment or any of its extensions, e.g., `tabularx`. The commands `\multirow` and `\multicolumn` allow table elements to span multiple rows and columns.

| Position    |        |                  |
|-------------|--------|------------------|
| Group       | Abbrev | Name             |
| Goalkeeper  | GK     | Paul Robinson    |
| Defenders   | LB     | Lucas Radebe     |
|             | DC     | Michael Duburrry |
|             | DC     | Dominic Matteo   |
|             | RB     | Didier Domi      |
| Midfielders | MC     | David Batty      |
|             | MC     | Eirik Bakke      |
|             | MC     | Jody Morris      |
| Forward     | FW     | Jamie McMaster   |
| Strikers    | ST     | Alan Smith       |
|             | ST     | Mark Viduka      |

Table 3.3: Adapted example from the  $\text{\LaTeX}$  guide at <https://en.wikibooks.org/wiki/LaTeX/Tables>. This example uses rules specific to the `booktabs` package and employs the multi-row functionality of the `multirow` package.

### 3.4.3 Images

An image is added to a document via the `\includegraphics` command as shown in Figure 3.1. The `\subcaption` command can be used to reference subfigures, such as Figure 3.1a and 3.1b.

### 3.4.4 Mathematical Expressions

One of the original motivations for creating the  $\text{\TeX}$  system was the need for mathematical typesetting. To this day,  $\text{\LaTeX}$  is the preferred system to write math-heavy documents and a wide variety of functions aids the author in this task. A mathematical expression can be



(a) The ISTA logo at line width.



(b) The ISTA logo at half the line width.

Figure 3.1: The header logo at different sizes.

inserted inline as  $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$  outside of the text stream as

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

or as a numbered equation with

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}. \quad (3.1)$$

### 3.4.5 Pseudo Code

The presentation of algorithms can be achieved with various packages; the most popular are `algorithmic`, `algorithm2e`, `algorithmicx`, or `algpseudocode`. An overview is given at <https://tex.stackexchange.com/questions/229355>. An example of the use of the `algorithm2e` package is given with Algorithm 3.1.

---

#### Algorithm 3.1: Gauss-Seidel

---

**Input:** A scalar  $\epsilon$ , a matrix  $\mathbf{A} = (a_{ij})$ , a vector  $\vec{b}$ , and an initial vector  $\vec{x}^{(0)}$

**Output:**  $\vec{x}^{(n)}$  with  $\mathbf{A}\vec{x}^{(n)} \approx \vec{b}$

```

1 for  $k \leftarrow 1$  to maximum iterations do
2   for  $i \leftarrow 1$  to  $n$  do
3      $x_i^{(k)} = \frac{1}{a_{ii}} (b_i - \sum_{j < i} a_{ij} x_j^{(k)} - \sum_{j > i} a_{ij} x_j^{(k-1)})$ ;
4   end
5   if  $|\vec{x}^{(k)} - \vec{x}^{(k-1)}| < \epsilon$  then
6     break for;
7   end
8 end
9 return  $\vec{x}^{(k)}$ ;

```

---

## 3.5 Bibliography

The referencing of prior work is a fundamental requirement of academic writing and is well supported by L<sup>A</sup>T<sub>E</sub>X. The BIB<sub>T</sub>E<sub>X</sub> reference management software is the most commonly used system for this purpose. Using the `\cite` command, it is possible to reference entries in a `.bib` file out of the text stream, e.g., as [Tur36]. The generation of the formatted bibliography needs a separate execution of `bibtex.exe` (see Table 3.2).

## 3.6 Table of Contents

The table of contents is automatically built by successive runs of the compilation, e.g., of `pdflatex.exe`. The command `\setsecnumdepth` allows the specification of the depth of the table of contents and additional entries can be added to the table of contents using `\addcontentsline`. The starred versions of the sectioning commands, i.e., `\chapter*`, `\section*`, etc., remove the corresponding entry from the table of contents.

## 3.7 Acronyms / Glossary / Index

The list of acronyms, the glossary, and the index need to be built with a separate execution of `makeindex` (see Table 3.2). Acronyms have to be specified with `\newacronym` while glossary entries use `\newglossaryentry`. Both are then used in the document content with one of the variants of `\gls`, such as `\Gls`, `\glspl`, or `\Glspl`. Index items are simply generated by placing `\index{<entry>}` next to all the words that correspond to the index entry `<entry>`. Note that many enhancements exist for these functionalities and the documentation of the `makeindex` and the `glossaries` packages should be consulted.

## 3.8 Tips

Since  $\text{\TeX}$  and its successors do not employ a What You See Is What You Get (WYSIWYG) editing scheme, several guidelines improve the readability of the source content:

- Each sentence in the source text should start with a new line. This helps not only the user navigate through the text but also enables revision control systems (e.g. Subversion (SVN), Git) to show the exact changes authored by different users. Paragraphs are separated by one (or more) empty lines.
- Environments, which are defined by a matching pair of `\begin{name}` and `\end{name}`, can be indented by whitespace to show their hierarchical structure.
- In most cases, the explicit use of whitespace (e.g. by adding `\hspace{4em}` or `\vspace{1.5cm}`) violates typographic guidelines and rules. Explicit formatting should only be employed as a last resort and, most likely, better ways to achieve the desired layout can be found by a quick web search.
- The use of bold or italic text is generally not supported by typographic considerations and the semantically meaningful `\emph{...}` should be used.

The predominant application of the  $\text{\LaTeX}$  system is the generation of PDF files via the `PDF $\text{\LaTeX}$`  binaries. In the current version of `PDF $\text{\LaTeX}$` , it is possible that absolute file paths and user account names are embedded in the final PDF document. While this poses only a minor security issue for all documents, it is highly problematic for double-blind reviews. The process shown in Table 3.4 can be employed to strip all private information from the final PDF document.

| Command                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 Rename the PDF document <code>final.pdf</code> to <code>final.ps</code> .                                                                                                                                                                                                                                                                                                                         |
| 2 Execute the following command:<br><pre>ps2pdf -dPDFSETTINGS#/prepress ^<br/>-dCompatibilityLevel#1.4 ^<br/>-dAutoFilterColorImages#false ^<br/>-dAutoFilterGrayImages#false ^<br/>-dColorImageFilter#/FlateEncode ^<br/>-dGrayImageFilter#/FlateEncode ^<br/>-dMonoImageFilter#/FlateEncode ^<br/>-dDownsampleColorImages#false ^<br/>-dDownsampleGrayImages#false ^<br/>final.ps final.pdf</pre> |
| On Unix-based systems, replace # with = and ^ with \.                                                                                                                                                                                                                                                                                                                                               |

Table 3.4: Anonymization of PDF documents.

## 3.9 Resources

### 3.9.1 Useful Links

In the following, a listing of useful web resources is given.

<https://en.wikibooks.org/wiki/LaTeX> An extensive wiki-based guide to L<sup>A</sup>T<sub>E</sub>X.

<http://www.tex.ac.uk/faq> A (huge) set of Frequently Asked Questions (FAQ) about T<sub>E</sub>X and L<sup>A</sup>T<sub>E</sub>X.

<https://tex.stackexchange.com/> The definitive user forum for non-trivial L<sup>A</sup>T<sub>E</sub>X-related questions and answers.

### 3.9.2 Comprehensive TeX Archive Network (CTAN)

The CTAN is the official repository for all T<sub>E</sub>X-related material. It can be accessed via <https://www.ctan.org/> and hosts (among other things) a huge variety of packages that provide extended functionality for T<sub>E</sub>X and its successors. Note that most packages contain PDF documentation that can be directly accessed via CTAN.

In the following, a short, non-exhaustive list of relevant CTAN-hosted packages are given together with their relative path.

**algorithm2e** Functionality for writing pseudo code.

**amsmath** Enhanced functionality for typesetting mathematical expressions.

**amssymb** Provides a multitude of mathematical symbols.

**booktabs** Improved typesetting of tables.

**enumitem** Control over the layout of lists (`itemize`, `enumerate`, `description`).

**fontenc** Determines font encoding of the output.

**glossaries** Create glossaries and lists of acronyms.



**graphicx** Insert images into the document.

**inputenc** Determines encoding of the input.

**l2tabu** A description of bad practices when using  $\text{\LaTeX}$ .

**mathtools** Further extension of mathematical typesetting.

**memoir** The document class upon which the `istaustriathesis` document class is based.

**multirow** Allows table elements to span several rows.

**pgfplots** Function plot drawings.

**pgf/TikZ** Creating graphics inside  $\text{\LaTeX}$  documents.

**subcaption** Allows the use of subfigures and enables their referencing.

**symbols/comprehensive** A listing of around 5000 symbols that can be used with  $\text{\LaTeX}$ .

**voss-mathmode** A comprehensive overview of typesetting mathematics in  $\text{\LaTeX}$ .

**xcolor** Allows the definition and use of colors.



# Bibliography

- [Tur36] Alan Mathison Turing. On computable numbers, with an application to the Entscheidungsproblem. *J. of Math*, 58:345–363, 1936.



# APPENDIX **A**

## **Example Appendix Chapter**



# List of Todos

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |      |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| Summarize the gist of your thesis in one page. . . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | vii  |
| Mention grants if you were funded. Consider Scientific Service Units (SSUs) at ISTA and other staff if you got their support. . . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | viii |
| Include paragraph of text summarizing your educational background, your academic profile (e.g. relevant research projects or work experience, list of publications), to give readers a snapshot idea of you as a person/academic. It is not meant to be a CV in the sense that it is not intended for job applications. This means things like your email address, your nationality, your current address, etc., should be left out.                                                                                                                                                                                                                                                                                                                                                                                          | ix   |
| The following is an entirely fictional blurb to give you an idea:<br>Jane Doe completed a BSc in Natural Sciences at the University of Cambridge and an MSc in Cell Biology at the University of Edinburgh before joining ISTA in September 2016. Her main research interests include evolution of genetic traits and environmental pressures on evolution. She worked on the research project “Mathematical Modeling of Evolutionary Constraints” with the Abel group at the University of Vienna in the summer of 2014, and published these results in the high-impact journal Evolutionary Biology. During her PhD studies, Jane also presented her research results at the EvoBIO conference in Parma in 2018 and helped develop the tool EVOBOT for modeling evolutionary constraints, currently hosted by ISTA. . . . . | ix   |
| Specify all co-authors and collaborator contributions to this thesis. See the Guidelines for Thesis Submission for details. . . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | x    |
| List all publications that appear in this thesis. . . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | x    |
| Use the <code>\bibentry</code> command to produce a full citation: . . . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | x    |
| Enter your text here. . . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 1    |
| Enter your text here. . . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 3    |